Autonomous Heart Rate Detection

Powell Burke
College of Nanotechnology, Science, and Engineering

Professor Hany Elgala
College of Nanotechnology, Science, and Engineering

UNIVERSITY AT ALBANY STATE UNIVERSITY OF NEW YORK

UALBANY SHOWCASE

INTRODUCTION

- Machine learning can improve disease detection and treatment while alleviating burden on medical professionals
- In the medical domain, video may give more detailed insight than still images
- ML models process videos of the same thing happening at different speeds differently. This can impair model performance on something like cardiac imaging where the heart rate can vary
- For cardiac imaging specifically, if the heart rate can be detected from a video, the video can be normalized to give the model a uniform representation of anatomical motion across varying rates of motion

BACKGROUND

- The specific problem of varying heart rate affecting AI performance in the medical domain was experience when working on the EASyExam Capstone Project
- In this project, the heart rate was detected using a method based on machine learning combined with signals processing and exploiting the similarity of cardiac cycles to one another
- This method will be tested here on the open source Stanford Echonet Dynamic Cardiac Ultrasound dataset
- Methods using end to end learning in a deep neural network and classical mulitple linear regression will be tested for heart rate detection on the same dataset to be compared

METHODS

Three methods were used to detect heart rate:

- End to end learning over CNN + Temporal Transformer Model
  - A Resnet-18 CNN was modified to take a 2 channel optical flow input and output a 512 dim embedding vector per frame
  - 100 frames per video were processed with each embedding becoming a token in a temporal transformer layer and with a CLS token prepended
  - The CLS token projects linearly to a heart rate prediction layer
- CNN per frame + Singal processing over embedding outputs
  - The same Resnet-18 structure was utilized but trained to maximize similarity between frame representations at the same period per video, and minimize similarity at frames at a phase offset from the video
  - To predict HR, embeddings are found for video frames using the trained CNN, then self similarity between all frames is computed. The average similarity at each interval of frames constructs a periodicity signal. Peaks in this signal represent the period of the heart rate and its harmonics
- MLP with optical flow-based feature inputs
  - Optical flow was found for up to 400 frames per video, and then per pixel features from all optical flow frames were found as: mean, p90, p95, p99, p99.9, and stddev
  - These features were used to train a multiple linear regression model to predict heart rate

Results	MAE (BPM)	Median AE (BPM)	P90 AE (BPM)
End to end	6.95	4.65	18.31
CNN + signal processing	4.07	0.96	8.02
MLP	10.07	7.96	21.23

CONCLUSION

All methods showed promise, with CNN + signal processing producing the best results.

These methods could be expanded or changed to detect period in other video formats where periodicity exists. This could be in the medical domain specifically or outside of it.

The strength of the CNN + signal processing method supports that ML may best be utilized when the scope of a problem is narrowed and the architecture is tailored to address the specific problem at hand. As model capabilities grow, there is increased incentive to utilize end to end learning and to absolve human engineering responsibility of finding a solution to the model. This work suggests that this may not yet yield the optimal solution.

REFERENCES

Stanford AIMI Center. (n.d.). *EchoNet-Dynamic Cardiac Ultrasound* [Dataset]. https://doi.org/10.71718/yqp5-y078

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).